

**18** At time  $t = 0$  hours a tank is full of water.

Water leaks from the tank.

At the end of every hour there is 2% less water in the tank than at the start of the hour.

The volume of water, in litres, in the tank at time  $t$  hours is  $V_t$

Given that

$$\begin{aligned} V_0 &= 2000 \\ V_{t+1} &= kV_t \end{aligned}$$

write down the value of  $k$ .

$$k = \dots\dots\dots$$